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# Transient Dynamics Simulation Platform for PDC, Roller Cone, and Hybrid Bits Utilizing Discrete Geometry

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## Abstract

Studies have shown that both the design of individual PDC cutters and their arrangement on a PDC bit significantly influence a bit's mechanical efficiency, ROP, and dynamic behavior. This paper reintroduces a simulation platform for PDC, roller cone, and hybrid drill bits that enables the analysis, optimization, and prediction of bit performance. The high-fidelity bit dynamics application was custom- built from the ground up to model bit-rock interaction and drilling dynamics with high efficiency. The application is implemented as object-oriented C++ code and employs a 3D triangulated surface mesh to accurately represent complex topology.

A toolkit was developed to automate the extraction of tessellation data from 3D CAD models and export it to a structured XML format for use in the software. Unlike using unstructured formats such as STLs, this process preserves both the hierarchical structure and surface designation information in the CAD model and transfers it to the software. This allows for detailed component and sub- component analysis, enabling the simulator to provide insights beyond the reach of physical testing. For instance, the geometric interactions between the bit and rock can be analyzed to compute contact areas, volumes of removed material, and forces at the sub- cutter level (e.g., chamfers, faces). As a demonstration, the WOB is decomposed into contributions from the cutter chamfers and cutter faces, providing a level of resolution unattainable through laboratory or field measurements.

To assess the performance and behavior of proprietary drill bit designs, laboratory tests were conducted in Pau, France. These results were used to validate the computer simulations. ROP and RPM from the lab were input into the model, while WOB and TOB were compared as outputs. To improve quantitative agreement between simulation and lab measurements, a novel optimization framework was introduced. This method uses post-processed results as a global fitting to align simulation outputs with measured values. The method drastically reduces optimization time by avoiding repeated simulation runs for each parameter update. The optimized fit is then implemented into the simulation framework to improve the predictive capability for new formations.

# Introduction

## Background

Studies have shown that drill bit design significantly influences the mechanical efficiency, rate of penetration (ROP), bit durability, system steerability, dynamic behavior, and borehole quality (Pastusek et al., 1992; Menand et al., 2004; Richard et al., 2007; Pelfrene et al., 2011; Jain et al., 2014; Cuillier et al., 2017; Pelfrene et al., 2019). The industry has sought to optimize the balance between these often competing objectives. Given the complex geometry present in both the design of drill bits and the bit-rock interaction, it was natural to build computer models to gain insights into how drill bit design affects its behavior both independently and as part of the drilling system.

Modeling drill bit performance has a long history in the industry. Equation-based methods were developed to predict forces (Briggs, 1968) and torques and predict ROP (Winters et al., 1987). The modern age of computer-based models began in the late 1980's. Dekun Ma, in a series of papers, developed the mathematics for roller cone bits and a computer model based on them (Ma and Yang, 1985; Ma and Azar, 1985; (Ma et al., 1995). Static models for polycrystalline diamond compact (PDC) bits Warren and Sinor, 1986; Glowka, 1987; Menand et al., 2002; Cuillier et al., 2017) emerged and progressed to kinematic (Behr et al., 1991) and then dynamic (Langeveld, 1992; Hanson and Hansen, 1995; Mahjoub et al., 2025). The model of Hanson and Hansen evolved to include roller cones (Dykstra et al., 2001) and contact surfaces for gauge pads and blades on PDC bits (Spencer et al., 2013). These models established a foundation upon which future developments were built, with some industry models still in use today tracing their origins—directly or indirectly— back to them.

The early computer models used analytical geometry. While this is computationally efficient, it limits the topography that can be easily represented and requires custom algorithms for each shape. Due to limited computational resources and the complexity of implementing analytical algorithms for most bit features, these models primarily focused on the cutting structure. When features involved complex geometries that were not amenable to analytical solutions, geometric approximations were employed.

Advances in computing power eventually allowed discrete geometry to be used (Endres, 2007; Pelfrene et al., 2019; Matthews et al., 2021). Using discretization allows arbitrary geometry to be represented and generalized algorithms to be utilized. This allows for modeling the complex geometry of features such as the bit body, blades, and shaped cutters with more precision than analytical geometry.

## Drill Bit Models

The process of rock removal has been studied using several approaches, with two main ones playing important roles: single cutter and full bit. Each of these offers unique insights and trade-offs.

The single cutting element approach studies the process of how the rock deforms, is crushed, and removed in relation to both the formation and the geometry of the cutting element. In the case of atmospheric pressure conditions, this research attempts to quantify the forces from cutting and friction and the spalling (indentation oversize) that occurs (Glowka, 1987; Detournay and Defourny, 1992). In the case of a pressurized environment, the research studies the forces while also accounting for effects such as hydraulic hold down, formation failure criteria, elastic-plastic effects of the rock, and crushed material flow (Ledgerwood, 2007; Gamwo et al., 2010).

To model the entire drill bit, simplifications are required to calculate the forces resulting from rock removal. These models are largely an exercise in computational geometry as the geometric calculations account for most of the expense. As a result, several assumptions are made to make the simulations practical.

1. Drill bits are treated as rigid bodies, except for roller cones which are allowed to rotate.
2. For dynamic simulation, wear is neglected. The lengths drilled by a simulation are too short to introduce significant wear.

3. Any interference with the bit and rock removes the rock. There is no elastic or plastic behavior and there is no minimum pressure or force threshold required to remove the rock.
4. The volume of rock removed is the volume swept by the bit as it moves. Rebounding of the formation is not accounted for. However, in some cases, additional material is removed to account for spalling of the rock at atmospheric conditions.
5. Hydraulic effects such as hydrostatic pressure, cuttings removal (or lack thereof), or jet impact are not directly considered.

These assumptions must be treated through the use of the cutting models, also referred to as bit-rock interaction models, force laws, or physics models. The forces generated from cutting can be represented as:

$$F = F(R, S, \text{ bottom hole pressure, damping}) \tag{1}$$

$$R = R(\text{Poisson's ratio, modulus of elasticity, CCS/UCS, coefficient of friction}) \tag{2}$$

$$S = S \text{ (velocity, area of cut, volume of cut, edge lengths along cut)} \tag{3}$$

Where $F$ is the cutting force, $R$ are formation dependent parameters, and $S$ are values calculated by the simulation.

It is common for bit-rock interaction models to be hybrid physics and data driven models (Endres, 2007; Matthews et al., 2021). In this approach, a model of known physical behavior is created that contains parameters that can be adjusted to refine the model's predictions. The parameters are tuned through an optimization process that fits the simulation results to data taken from the laboratory or field. The parameters account for unknown physics or physics that would be too expensive to calculate for a drill bit model.

### Current Work

The present work builds upon the High-Fidelity Bit Dynamics Software introduced by Endres (2007), developed to model and predict transient drill bit dynamics. The objective of this study was to integrate that application into a workflow for assessing drill bit designs with respect to their dynamic behavior. Achieving this required addressing two key challenges. First, the 3D drill bit models had to be discretized into triangulated surface meshes and exported to a proprietary format. Second, the bit-rock interaction model required optimization. In this paper, we discuss several of the principal issues encountered and the techniques applied to resolve them. Although the software is capable of modeling PDC, roller-cone, and hybrid drill bits, we focus here on PDC bits to demonstrate the process. All theory developed here for PDC bits also applies to the other bit types.

## High-Fidelity Bit Dynamics Software Introduction

The High-Fidelity Bit Dynamics Software (Endres, 2007) introduced several firsts in regard to modeling drill bit behavior:

1. The use of discretized geometry in the form of triangulated surface meshes to represent the bit and rock surface. This enables high geometric fidelity for arbitrary surface topology.
2. Object-oriented, separation-of-interests code design that creates an adaptable platform. This makes it easy to accommodate new bit designs, new bit features, and new bit-rock interaction models.
3. Representation of the entire bit. Previous models were limited to the cutting structure and a few other strategic elements.
4. Incorporation of PDC, roller cone, and hybrid bits into the initial design (see Fig. 1). Other software started as code to model one bit type and were later adapted to other bit types.

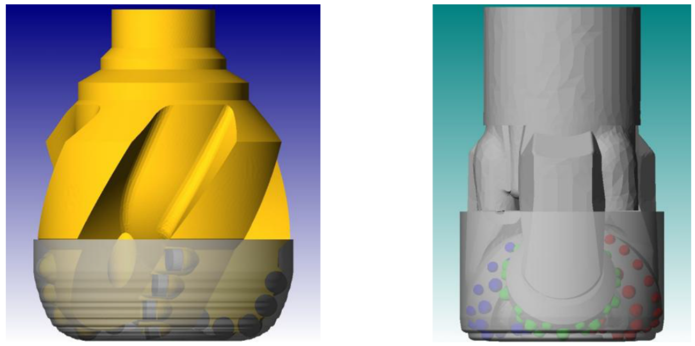

**Figure 1—Example PDC and roller cone simulations. From Endres (2007).**

The software allows defining arbitrary portions of a drill bit, referred to as *BITPARTs*, that govern modeling behavior. Each *BITPART* may include arbitrary geometry and bit-rock interaction models. *BITPARTs* are defined according to the desired behavior and modeling requirements and are organized hierarchically. For instance, a *CUTTER* is composed of a *CUTTERFACE*, *CUTTERCHAMFER*, and *CUTTERBODY*. This hierarchical structure provides flexibility, allowing different bit-rock interaction models to be applied at the chamfer and face levels while still aggregating the results to compute the overall cutter response.

## Discretization and Preprocessing of the Bit Geometry

The flexibility and generality of the software requires that geometrical features be associated with their respective *BITPART*. Common file formats, such as STLs (stereolithography), contain unstructured data; the association of the tessellation with the original geometry is lost. Furthermore, the software requires additional information about the drill bit such as its mass, moments of inertia, and local coordinate systems. To facilitate the transfer of both the additional properties, and the structured geometric data, a proprietary file format is used.

Modern drill bits are designed in advanced 3D computer-aided design (CAD) packages. It is common for these software applications to already include algorithms for generating surface tessellations. However, these tessellations do not inherently retain information about the parent geometry. To associate individual mesh elements with the corresponding surfaces in the CAD model, a plugin was created. The toolkit enables users to tag various parts (surfaces) in the CAD environment, group them into categories such as cutter face, cutter chamfer, cutter body, gauge pad, bit body, etc. and generate tessellated output that maintains the relationship between the discretized geometry and bit feature. Notably, each triangle in the exported mesh is traceable to a specific tagged surface, enabling high-fidelity mapping between CAD and simulation. The plugin also maintains the hierarchical nature of the bit (for example, a cutter face belongs to a cutter, and the cutter belongs to the bit). This information is then exported to a structured Extensible Markup Language (XML) format.

The toolkit simplifies and automates the process of generating the bit mesh. It includes features to assist in tagging surfaces, calculates coordinate systems, numbers the cutters, calculates moments of inertia and mass, creates the tessellation, and exports the results. It also allows for controlling the discretization size based on different features of the bit. This allows a finer mesh to be created on the cutter chamfers, where precision is required, and a coarser mesh on the bit body where less detail is required, as seen in Fig. 2.

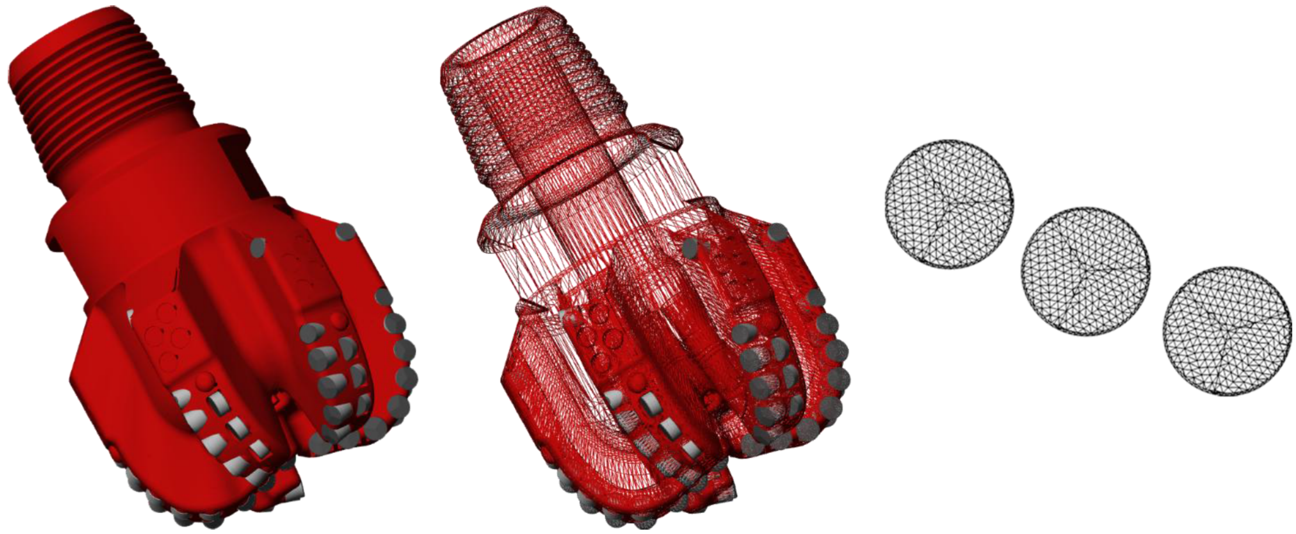

**Figure 2—Discretizing the bit. On the left is the original bit. In the center, the bit has been discretized with a triangulated surface mesh. It can be seen that the body of the bit has a courser mesh than those of the cutter face and chamfer, as seen in close up at the right.**

## Laboratory Tests and Software Optimization

### Testing Apparatus

Laboratory tests were conducted in Pau, France on equipment described in (Gerbaud et al., 2006; Mahjoub et al., 2025). The test rig (see Fig. 3) measures depth, weight on bit (WOB), rotation speed, torque on bit (TOB), and accelerations (on the drill string, approximately 1 meter from the bit). ROP is calculated in post processing and not directly measured.

**Figure 3—Drilling test bench. From Stoxreiter et al. (2019).**

Tests were performed on a hard sandstone under both atmospheric and pressurized conditions. To replicate downhole environments, a pressure vessel capable of simulating subsurface conditions was used (see Fig. 4). Pressurized tests were conducted at 10 MPa. The sandstone has an unconfined compressive strength (UCS) of 166 MPa, a confined compressive strength (CCS) of 263 MPa at 10 MPa pressure, and an internal friction angle of 44 degrees.

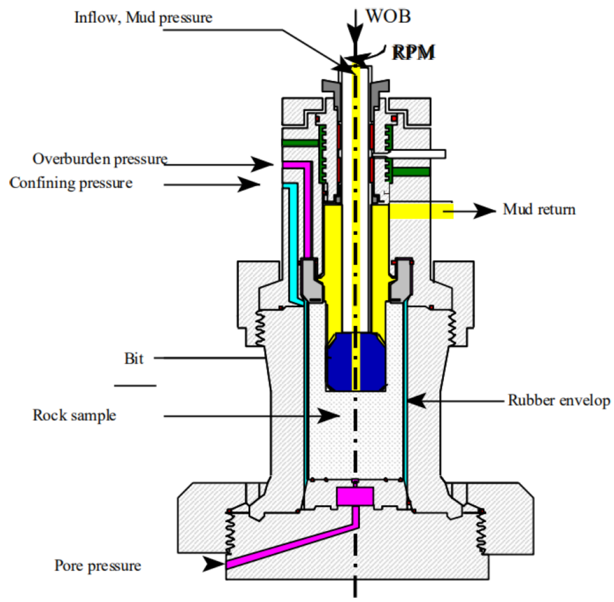


**Figure 4—Schematic of the laboratory testing equipment used to simulate downhole pressurized conditions. From Stoxreiter et al. (2019).**

Ten total tests were selected for use, 5 atmospheric and 5 pressurized. The tests did not follow the same test protocol. In addition, control issues contributed to differences between them. Moreover, the WOB is indirectly controlled by the hydraulic system and thus introduces additional error and sometimes drifts. Therefore, we opted not to directly compare tests to one another.

## Drill Bit Description

In total, 5 drill bits were used. The base design for each was the same, an 8.5″, 5-bladed PDC bit. The cutters were 16 mm with 0.23 mm chamfers and back rakes that ranged from 11 degrees at cutter number 1 to 27 degrees on the shoulder. An example of bit used can be seen in Fig. 5.

**Figure 5—Bit used in laboratory tests. It is an 8.5″ PDC bit with 16 mm cutters.**

## Preparing Simulations

We built inputs for the High-Fidelity Bit Dynamics Software with the goal of reconstructing the laboratory tests. To this end, we began by analyzing and processing the laboratory data, aiming to extract *zones of interest* to reconstruct.

The WOB data was processed through a lowpass filter, with cutoff frequencies tailored to observations from each individual test. These vary between 1.25 Hz, 2 Hz and 6 Hz. The output was fed to a signal segmentation algorithm (Radhakrishnan et al., 1991; Endres, 2015), segmenting the data into several zones where the signal was considered statistically constant. The process was repeated for the rotary speed signal, and we took the intersection of both sets. A final manual pass was made by a subject matter expert to merge and discard zones to satisfy our set constraints. Fig. 6 contains an example showcasing the manual zone operations performed for one of the atmospheric tests.

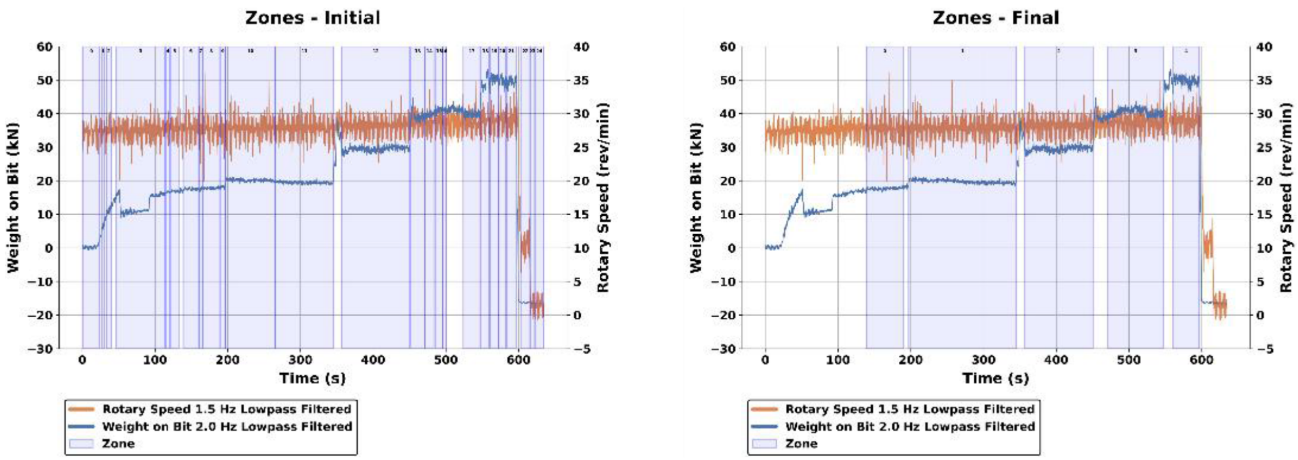


Figure 6—Example of zone operations for an atmospheric test.

From the final set of processed zones, we extracted the per-zone average of lowpass-filtered ROP and rotary speed. Fig. 7 shows the final zone average values for the same example as Fig. 6.

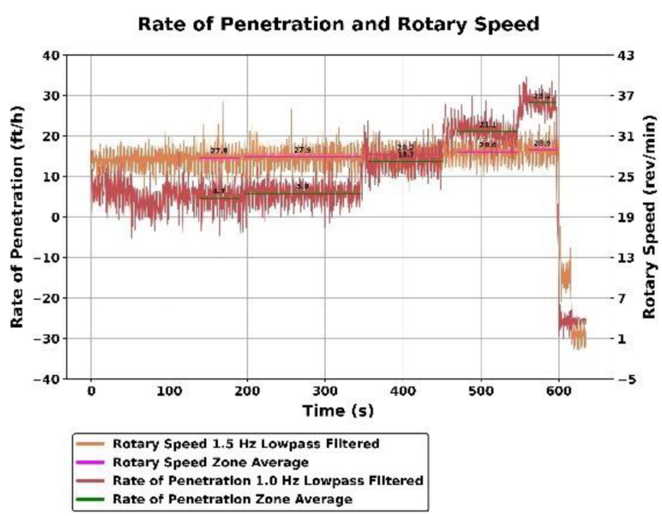


Figure 7—Example of zone average values for an atmospheric test.

To recreate each test in the High-Fidelity Bit Dynamics Software, we created a simulation that starts with two zones. The first spuds the bit into the rock, running for 5 revolutions with an ROP of 10 m/h and a rotary speed of 120 rpm. The second zone transitions out of the spudding, running for 2 revolutions with an ROP of 1 m/h. The rotary speed for this zone matches the first zone that comes after, guided by the laboratory data.

We then dynamically created the remaining zones with the information coming from the lab data analysis. Staying on the same example, from the data observed in Fig. 7 we create the zones detailed in Table 1. Each zone runs for a total of 19 revolutions, with a transition of 0.5 seconds between consecutive zones.

**Table 1—Example of zones created for a simulation to recreate an atmospheric test.**

| Zone | Revolutions | ROP (ft/h) | Rotary Speed (rpm) |
|---|---|---|---|
| Spud | 5 | 32.8084 (10 m/h) | 120 |
| Transition out of Spud | 2 | 3.28084 (1 m/h) | 27.778 |
| 1 | 19 | 4.663 | 27.778 |
| 2 | 19 | 5.856 | 27.890 |
| 3 | 19 | 13.746 | 28.247 |
| 4 | 19 | 21.107 | 28.641 |
| 5 | 19 | 28.183 | 28.926 |

## Simulation Results

The created plugin, described in Subsection *Discretization and Preprocessing of the Bit Geometry*, was used to recreate the bit XML files. Then, after creating the simulation input files according to the previous section, we ran 10 simulations corresponding to the laboratory tests. From the simulation results, we extract the zones of interest to match the laboratory data.

Fig. 8 shows the simulation results of the same test previously used as an example. First, we can observe that the number of zones extracted from the simulation matches the five seen in Fig. 6 and Fig. 7. Second, we show the comparison of TOB and WOB zone average values to the equivalent ones extracted from the lab data.

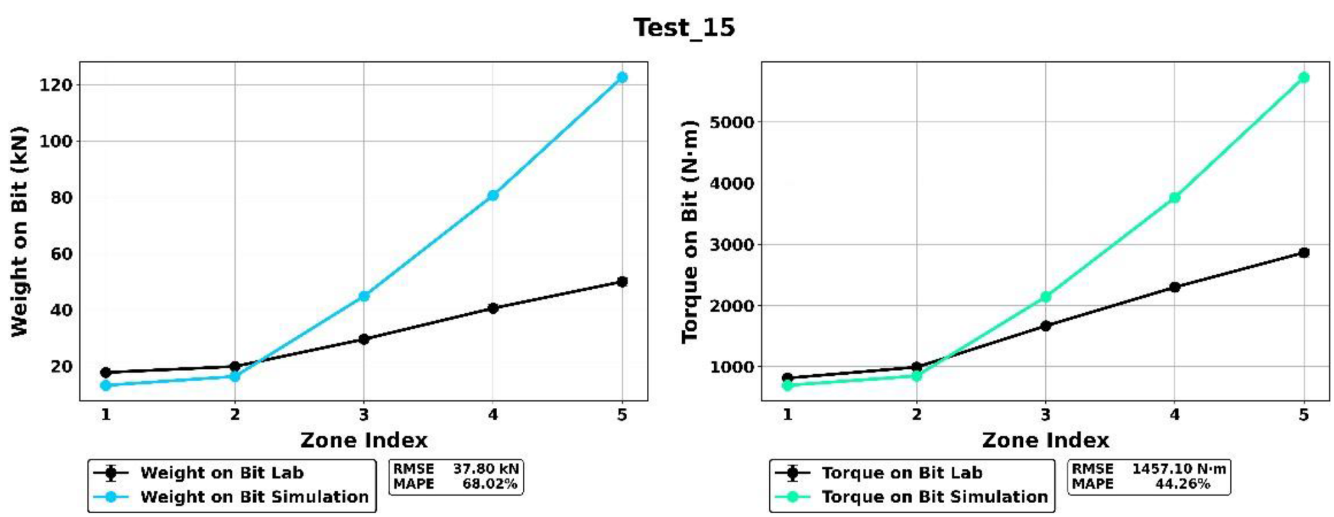


**Figure 8—Example of simulation results for an atmospheric test.**

## Simulation Optimization

From the plots in Fig. 8, we observe a nontrivial mismatch between laboratory and simulation data. This was expected as the software was not previously used for Rhune sandstone, and the formation type is a controlling parameter for the predictive ability of the software. A typical optimization follows the process below.

1. A new set of coefficients are generated (in this case, the coefficients are used in the bit-rock interaction model to predict forces).
2. The new coefficients are used in the simulation software, generating new values (in this case, WOB and TOB).
3. An objective function calculates a value representing the difference between the data and simulation values.

The process repeats, while attempting to minimize the objective function. The issue with this method is that for each iteration, a complete set of simulations must be run. This is an expensive process.

To reduce the time required to optimize the simulation's predictive power, a new method was developed. The method works by optimizing the existing simulation output to the laboratory data. A global fitting function is generated. This can then be implemented into the software. This is a post-hoc method, in the sense that its application occurs after simulations have been completed.

We begin by describing the necessary notation. We first define $[\cdot] := \{1,\ldots,\cdot\}$. We say a set of data $s^*$, representing a laboratory test, has $n_s \in \mathbb{N}$ zones. We concern ourselves, throughout this framework, with the *zone average data*. Thus, a set of laboratory data is a collection of WOB and TOB average values for each zone, which in notation is

$$s^\star = \left\{\left(\mathrm{WOB}^\star_{s_w}, \mathrm{TOB}^\star_{s_w}\right) : w \in [n_s]\right\} \in \mathbb{R}^{n_s \times 2}. \tag{4}$$

We now define the simulation $s$ that corresponds to $s^*$. First, we say that $s$ has $T_s \in \mathbb{N}$ timesteps. We define the set-valued function $\tau_s : [n_s] \rightrightarrows [T_s]$ that gives the set of timesteps belonging to each zone. We also say that our bit, shared by all simulations, has $C \in \mathbb{N}$ cutter parts, i.e. cutter faces and cutter chamfers.

The outputs of simulation $s$ are the forces $F_s = \left(F_{S_x}, F_{S_y}, F_{S_Z}\right) \in \mathbb{R}^3$ and their locations in the bit frame of reference $D_s = \left(D_{S_x}, D_{S_y}, D_{S_Z}\right) \in \mathbb{R}^3$. Notice then that for these, WOB is given by $F_{S_Z}$, and TOB by $(D_s \times F_s)_z$. Also, since these are in fact dependent on the cutter part and timestep, for ease of notation, we refer to them through functions $F_s, D_s : [C] \times [T] \to \mathbb{R}^3$.

It thus holds that the zone average values for simulation WOB and TOB are given by

$$\mathrm{WOB}_{s_w} = \frac{1}{|\tau_s(w)|} \sum_{t \in \tau_s(w)} \sum_{c=1}^{C} F_s(c,t)_z, \quad \mathrm{TOB}_{s_w} = \frac{1}{|\tau_s(w)|} \sum_{t \in \tau_s(w)} \sum_{c=1}^{C} \left[ D_s(c,t) \times F_s(c,t) \right]_z, \tag{5}$$

and the simulation $s$ is

$$s = \left\{\left(\mathrm{WOB}_{s_w}, \mathrm{TOB}_{s_w}\right) : w \in [n_s]\right\} \in \mathbb{R}^{n_s \times 2}. \tag{6}$$

We now detail the steps of the developed methodology. Note that we illustrate the steps for a generic set of simulations $S := \{S_i\}_{i=1}^{N}$ and accompanying laboratory data $S^* := \{S_i^*\}_{i=1}^{N}$, for some $N \in \mathbb{N}$ number of tests.

***Step 1: Coefficient Optimization.*** The framework begins with the selection of one or more tests to serve as the "training set", i.e. the data for which the coefficients will be optimized on. We refer to this through indexes $T \subseteq [N]$.

To reduce the mismatch between laboratory data and the corresponding simulations, we transform the simulation data with the objective of minimizing the final difference to the lab results. The transformation coefficients are designed to act on the forces through $F \mapsto a \circ F + b$, where $a = (a_{xy}, a_{xy}, a_z) \in \mathbb{R}^3$ and $b = (b_{xy}, b_{xy}, b_z) \in \mathbb{R}^3$. In essence, we have $(F_x, F_y, F_z) \mapsto (a_{xy}F_x + b_{xy}, a_{xy}F_y + b_{xy}, a_zF_z + b_z)$.

Our objective function, however, considers the final WOB and TOB to reduce their mismatch. We define the transformed zone averages

$$\mathrm{WOB}'_{s_w}(a,b) := \frac{1}{|\tau_s(w)|}\sum_{t\in\tau_s(w)}\sum_{c=1}^{c}\left[a\circ F_s(c,t)+b\right]_z, \quad \mathrm{TOB}'_{s_w}(a,b) := \frac{1}{|\tau_s(w)|}\sum_{t\in\tau_s(w)}\sum_{c=1}^{c}\left[D_s(c,t)\times\left[a\circ F_s(c,t)+b\right]\right]_z, \tag{7}$$

and via least squares, obtain the coefficients

$$(\alpha,\beta)\in \underset{\substack{(a,b)\in\mathbb{R}^3\times\mathbb{R}^3\\ a_x=a_y,b_x=b_y}}{\operatorname{argmin}} \sum_{j\in T}\sum_{w=1}^{ns_j}\left[\left[\mathrm{WOB}'_{s_{j_w}}(a,b)-\mathrm{WOB}^{\star}_{s_{j_w}}\right]^2+\left[\mathrm{TOB}'_{s_{j_w}}(a,b)-\mathrm{TOB}^{\star}_{s_{j_w}}\right]^2\right]. \tag{8}$$

Now, notice that

$$\mathrm{WOB}'_{s_w}(a,b) = \frac{1}{|\tau_s(w)|}\sum_{t\in\tau_s(w)}\sum_{c=1}^{c} a_z F_s(c,t)_z + b_z = a_z \mathrm{WOB}_{s_w} + b_z C, \tag{9}$$

and

$$\begin{aligned}\mathrm{TOB}'_{s_w}(a,b) &= \frac{1}{|\tau_s(w)|}\sum_{t\in\tau_s(w)}\sum_{c=1}^{c}\left[D_s(c,t)_x\left[a_{xy}F_s(c,t)_y+b_{xy}\right]-D_s(c,t)_y\left[a_{xy}F_s(c,t)_x+b_{xy}\right]\right]\\ &= a_{xy}\mathrm{TOB}_{s_w}+b_{xy}\frac{1}{|\tau_s(w)|}\sum_{t\in\tau_s(w)}\sum_{c=1}^{c}D_s(c,t)_x-D_s(c,t)_y =: a_{xy}\mathrm{TOB}_{s_w}+b_{xy}\overline{D_{s_w}}.\end{aligned} \tag{10}$$

This implies that we have two options on where the coefficients can act: either directly on the cutter part forces, or, with the care of adding the necessary extra terms, on the zone average WOB and TOB. We can also rewrite (8) as

$$(\alpha,\beta)\in \underset{\substack{(a,b)\in\mathbb{R}^3\times\mathbb{R}^3\\ a_x=a_y=:a_{xy},b_x=b_y=:b_{xy}}}{\operatorname{argmin}} \sum_{j\in T}\sum_{w=1}^{ns_j}\left[\left[a_z\mathrm{WOB}_{s_{j_w}}+b_zC-\mathrm{WOB}^{\star}_{s_{j_w}}\right]^2+\left[a_{xy}\mathrm{TOB}_{s_{j_w}}+b_{xy}\overline{D_{s_{j_w}}}-\mathrm{TOB}^{\star}_{s_{j_w}}\right]^2\right], \tag{11}$$

where $\mathrm{WOB}_{s_{j_w}}$, $C$, $\mathrm{WOB}^{*}_{s_{j_w}}$, $\mathrm{TOB}_{s_{j_w}}$, $\overline{D_{S_{J_W}}}$ and $\mathrm{TOB}^{*}_{s_{j_w}}$, are known or can be precalculated beforehand for all $j\in T$ and $w\in[n_j]$.

***Step 2: Coefficient Application.*** With the coefficients obtained in the previous step, we're now able to transform data that was not part of the training set. Rigorously, we transform the remaining simulations to

$$s'_i := \left\{\left(\mathrm{WOB}'_{s_{i_w}}(\alpha,\beta), \mathrm{TOB}'_{s_{i_w}}(\alpha,\beta)\right): w\in\left[ns_i\right]\right\} \quad \forall\, i\in[N]\setminus T. \tag{12}$$

Notice that based on the equivalencies shown in the previous step, we can transform the simulations either by applying the coefficients on the forces, and then calculating zone averages, or by first calculating zone averages and then applying the coefficients on the result, with the additional terms added as required.

***Step 3: Error Calculation.*** The results of each transformed simulation can then be compared to corresponding lab data. Recalling that for each simulation $\boldsymbol{s}$, all of $s$, $s^*$ and $s'$ have the same number of zones $n_s$, we do so via the following metrics:

- The *root mean squared error*, which returns errors in the same units as the data, defined as

$$\mathrm{RMSE}_k\left(s^{\star},s'\right) := \left[\frac{1}{n_s}\sum_{w=1}^{n_s}\left[s^{\star}_w(k)-s'_w(k)\right]^2\right]^{\frac{1}{2}}, \tag{13}$$

where $k\in\{1,2\}$ corresponds to WOB or TOB.

- The *mean absolute percentage error*, which returns percentual errors, defined as

$$\mathrm{MAPE}_k\left(s^{\star},s'\right) := \frac{1}{n_s}\sum_{w=1}^{n_s}\left|\frac{s^{\star}_w(k)-s'_w(k)}{s^{\star}_w(k)}\right|, \tag{14}$$

where again, $k \in \{1,2\}$ corresponds to WOB or TOB.

With the framework laid out, we now present results of its application to our laboratory data and corresponding simulations. Fig. 9 shows the plots and metrics obtained when selecting four atmospheric tests for the training set and calculating the results for the remaining single unseen test. Fig. 10 repeats this analysis for the pressurized tests. Each one of the plots shows the laboratory data, the original simulation data prior to coefficient optimization, and the final transformed simulation.

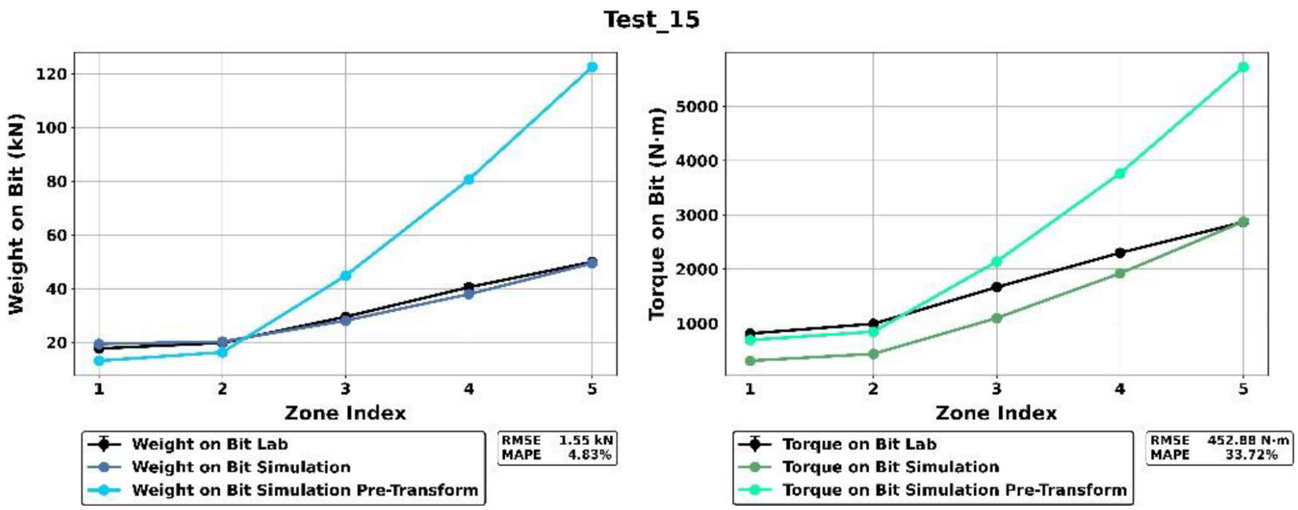


**Figure 9—Example of comparison plots and metrics for an atmospheric test.**

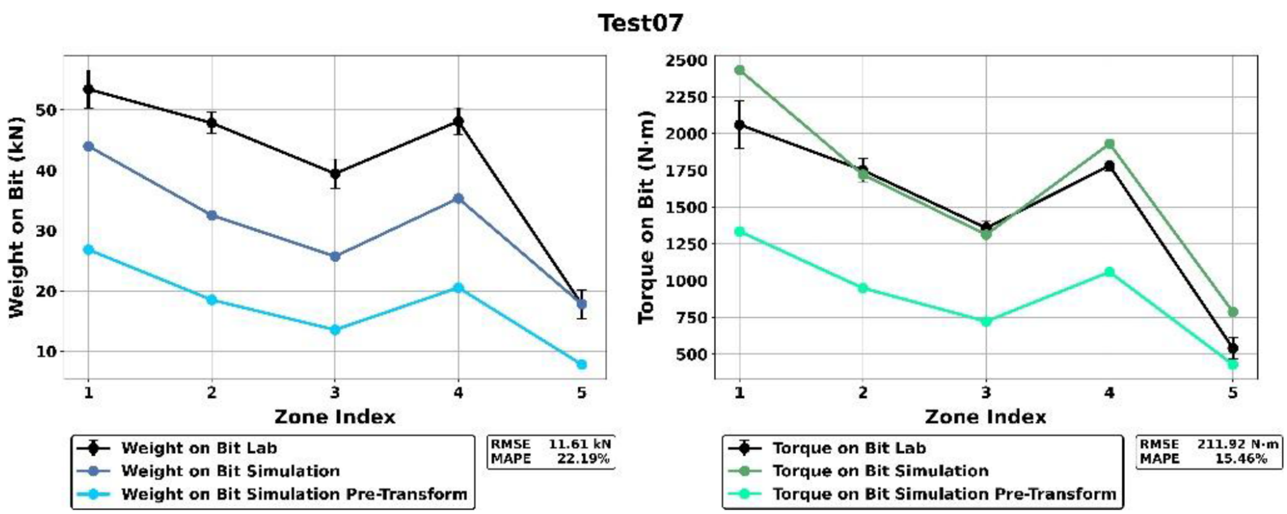


**Figure 10—Example of comparison plots and metrics for a pressurized test.**

We repeat this procedure in the fashion of *leave-one-group-out cross-validation*, iterating over the unseen test while optimizing the coefficients over the four remaining ones in each test collection. After iterating over all tests and transforming all data in this manner, we pool all data and compare to analogously pooled lab data. These results are shown in Fig. 11 and Fig. 12.

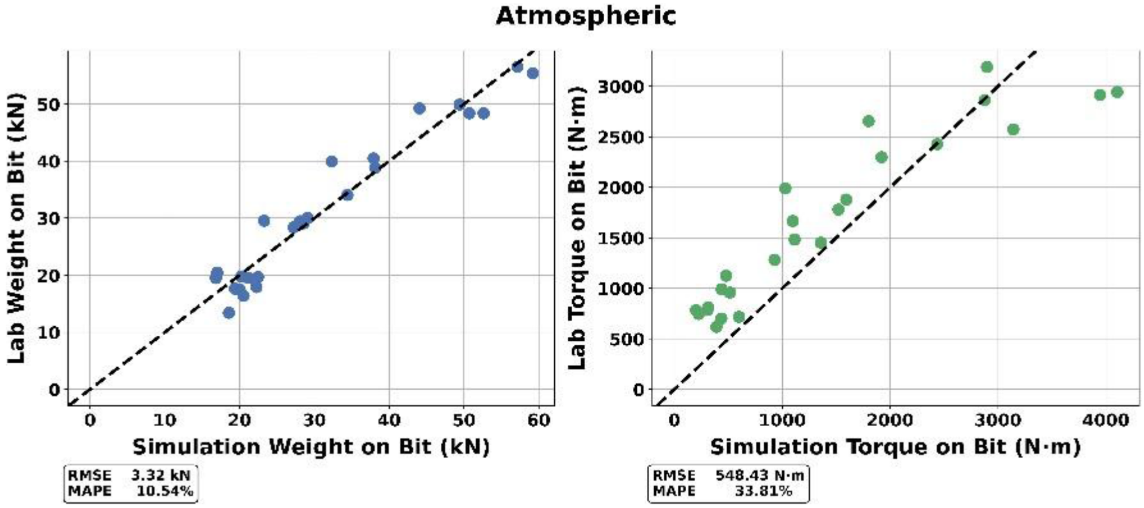


**Figure 11—Final plots and metrics for atmospheric tests.**

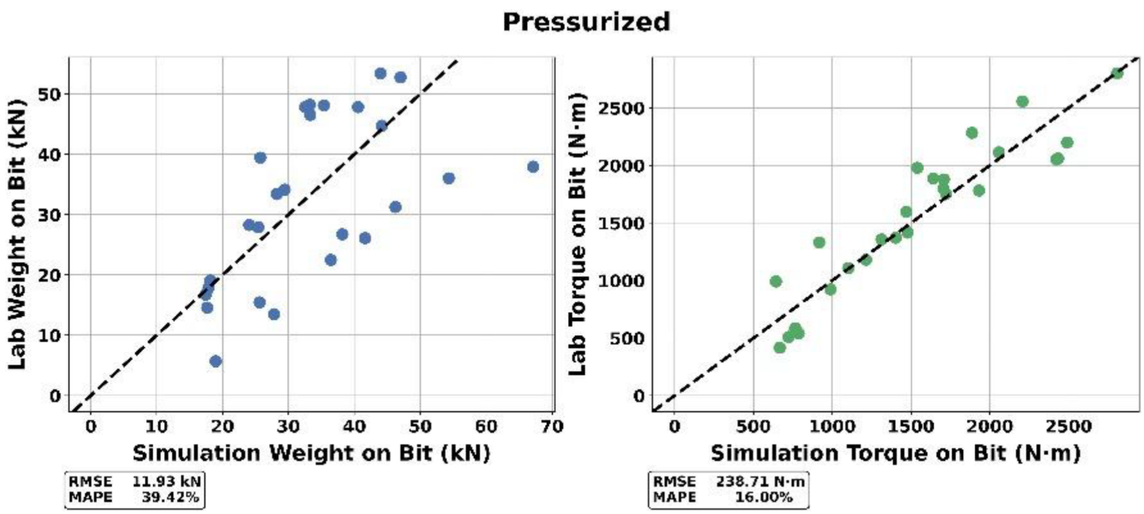


**Figure 12—Final plots and metrics for pressurized tests.**

Analyzing the above results, we can verify that the simulator better matched WOB data for the atmospheric tests, with an error of 10.54%, and TOB data for the pressurized tests, with an error of 16%. Curiously, we obtained better WOB than TOB results for the atmospheric data but observed the inverse behavior for pressurized ones.

The results allow us to verify that the simulator can be used as a tool to guide bit design in lieu of costly and lengthy laboratory tests. As an example, the simulator can aid in drastically reducing the initial search space of a bit-design optimization procedure, and laboratory tests can still be performed for a small set of final candidates to guarantee desired accuracy in the end result.

The optimization procedure also enables further insights into the laboratory tests. For instance, if selecting a collection of tests and training on all but one, poor metrics on the unseen test might reveal unknown or latent information. Subtle variations across tests, such as natural differences in the rock material, reduce model performance, in turn revealing the very existence of such differences.

## Online Optimization Application

Another benefit of the developed framework is that not only can we apply the coefficients *offline* (post simulation), as described in step 2 of the methodology, but also *online* (during the simulation).

After training, the obtained coefficients can be fed into the simulator itself, allowing for transformation of the data during the simulation, instead of after its end. The results analogous to those shown in Fig. 9, Fig. 10, Fig. 11 and Fig. 12, obtained by rerunning the same simulation with the online coefficient application, can be seen in Fig. 13, Fig. 14, Fig. 15 and Fig. 16.

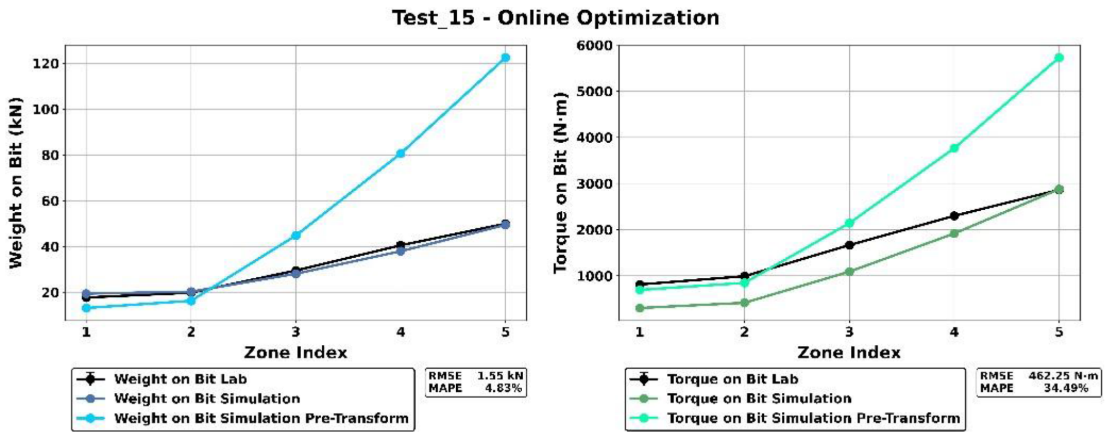


**Figure 13—Comparison plots and metrics for an atmospheric test with online coefficient application.**

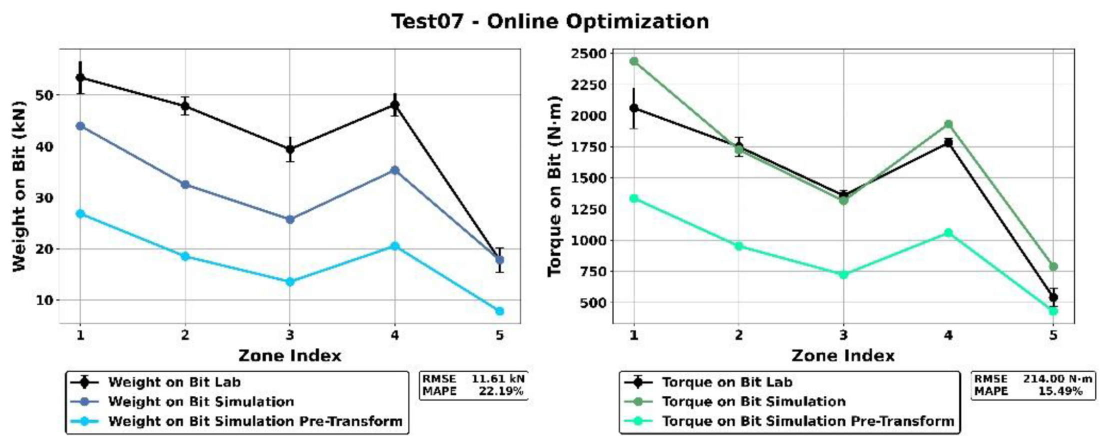


**Figure 14—Comparison plots and metrics for a pressurized test with online coefficient application.**

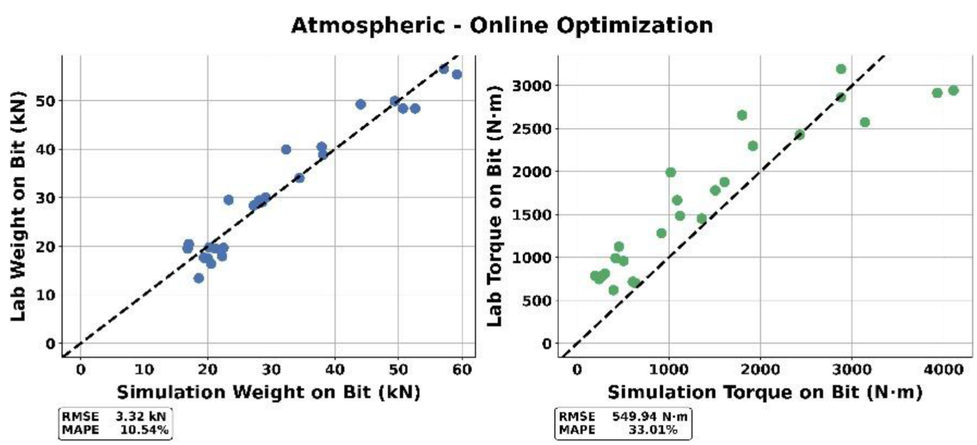


**Figure 15—Final plots and metrics for atmospheric tests with online coefficient application.**

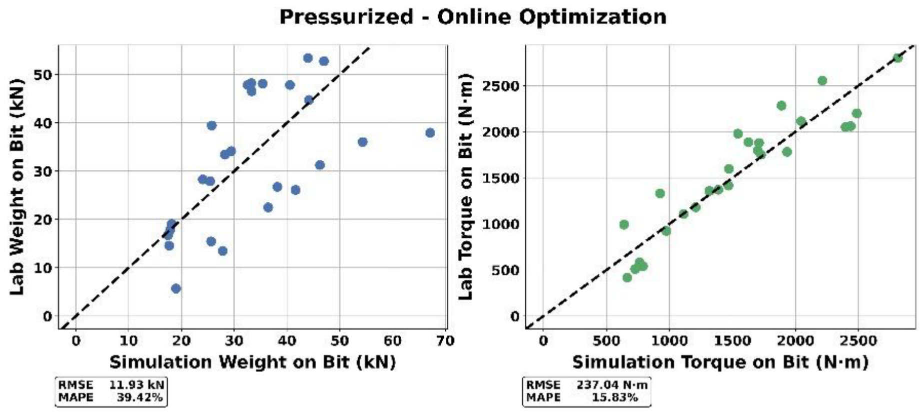


Figure 16—Final plots and metrics for pressurized tests with online coefficient application.

As expected, the results match those previously obtained through offline application. As the coefficients can be built into the simulator, we allow for automatic transformation without any post-simulation user effort. This also enables analyses of dynamics behavior, such as whirl, directly influenced by the lateral forces.

## Sub Cutter-Level Results

Among the advantages of the developed simulator lies the fact that it outputs results separately by "bit part." In the software, a cutter is further broken down into components for the cutter face, chamfer, and body. This allows us to analyze the drilling performance on the sub-cutter level which is not possible on laboratory tests.

To highlight this, we analyzed the results stemming from the laboratory tests described above. We now not only assess the total bit data, but also the results on the cutter faces and on the cutter chamfers. To this end, for each set of tests, we obtain from the simulator the isolated WOB data per cutter face, cutter chamfer, and whole bit. We add all 55 cutter faces to get the total WOB coming from cutter faces, and the same for the chamfers. This allows us to assess how much of the WOB is being carried by chamfers and faces independently. In turn, this gives an indication of the efficiency of the drilling.

We repeat the procedure described in the previous section of obtaining zone average values, now using the cutter faces and chamfers as the sources of data. With this, we can analyze WOB per bit part, shown in Fig. 17 and Fig. 18. From these figures we can first observe how the simulator follows our expected behavior: the sum of the WOB corresponding to cutter faces plus the sum of the WOB corresponding to the cutter chamfers is equal to the WOB of the bit.

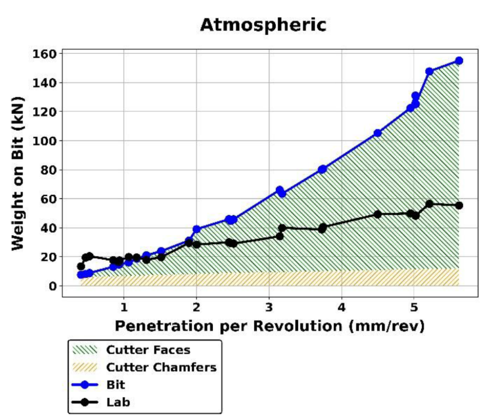


Figure 17—WOB per bit part for atmospheric data from pre- optimized simulation data.

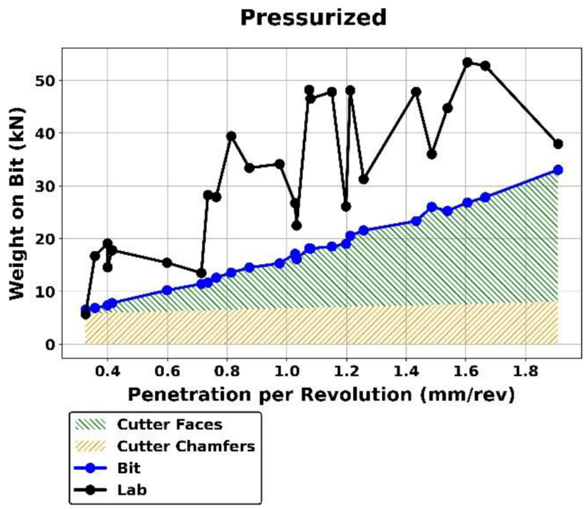


**Figure 18—WOB per bit part for pressurized data from pre- optimized simulation data.**

From these plots, we can see that at low penetration per revolution (PPR) the weight on bit is generated almost exclusively from the cutter chamfers. At high PPR rates, the cutter faces carry the majority of the weight. This matches the expected behavior. At low PPR rates, the depth is not enough to have engaged the cutter faces yet. After the formation begins to engage the faces, most of the additional cutting, as PPR increases, is handled by the faces.

Based on the description of the bit in Section *Drill Bit Description*, we can calculate that the faces on the cone cutters begin interacting with the formation at 0.226 mm in the cone and 0.223 mm at the nose. This agrees well with the plots where the contribution of the cutter faces goes to zero in this region.

These results were also obtained for the previously described online application of optimized coefficients, to better match the laboratory data. The analogous plots can be seen in Fig. 19 and Fig. 20.

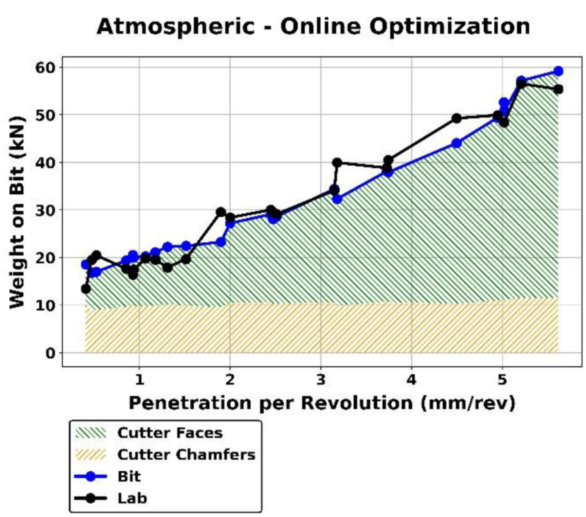


**Figure 19—WOB per bit part for atmospheric data from optimized simulation data.**

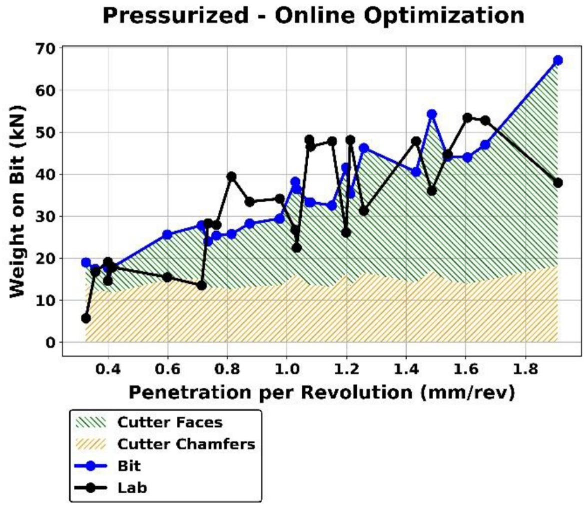


**Figure 20—WOB per bit part for pressurized data from optimized simulation data.**

# Conclusions

This paper presents the validation of a transient dynamics simulation platform designed to model the behavior of PDC, roller cone, and hybrid drill bits using high-resolution discrete geometry. The work industrialized the software by developing automated preprocessing to generate a proprietary file format that maintains the relationship between the discretized geometry and bit feature in the original CAD model.

To improve quantitative agreement between simulation and lab measurements in formations, a novel optimization framework was introduced. This post-processing method uses global fitting to align simulation outputs with measured WOB and TOB values using a low-dimensional polynomial transformation. The method drastically reduces optimization time by avoiding repeated simulation runs for each parameter update. The optimized model achieved a MAPE of approximately 10% for WOB in atmospheric tests and 16% for TOB in pressurized tests, demonstrating the simulator's accuracy. Although error levels were higher for WOB in pressurized tests and TOB in atmospheric ones, the simulation still captured the overall trends in behavior and showed promise for further refinement.

Through software design, component and sub-component associations are maintained from the bit design. This allows the simulator to provide component-level insights that are not available from physical testing. For instance, analysis of WOB contributions from cutter chamfers and faces revealed clear transitions in load distribution as a function of penetration per revolution. This provides valuable insights to bit design engineers.

Taken together, the results demonstrate that this simulation platform offers a robust and flexible environment for analyzing and optimizing drill bit performance across a range of designs and operational conditions. It reduces reliance on costly and time-consuming laboratory trials, offers predictive capability beyond experimental boundaries, and provides detailed data that can inform future bit design and development.

With the simulator now calibrated for the tests described, next steps to this work include analyzing bit-whirl behavior, enabled by the online application framework. We can also continue the tests by exploring more ROP and RPM zones through the simulator, without the need for new laboratory data collection. The predictive power of the simulator can also be further explored by running new tests, while pushing the boundaries of the optimization framework in testing how many simulations are in fact needed to model new formations.

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